

The Theory the Regressions Were Missing

Teaching note

Ohlson (1995), *Contemporary Accounting Research*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

A paradox in neoclassical valuation motivates the model

“the paper highlights the key role of accounting data when one tries to come to grips with an apparent paradox in neoclassical security valuation: the present value of expected dividends determines a firm’s value, yet the prediction of the dividend sequence is basically irrelevant if the underlying dividend policy is irrelevant. (Who wants to predict next year’s expected dividends when all dividend policies yield the same market value?)”

Ohlson (1995, pp. 681–682)

Last week, Easton and Zmijewski never stated the valuation model behind their regressions; their footnoted models hold only under certainty. **Ohlson resolves the paradox with accounting, supplying the theory the empirical literature had been assuming.**

Three assumptions generate the entire model

(A1) Present value of expected dividends, risk-neutral investors

$$P_t = \sum_{\tau=1}^{\infty} R_f^{-\tau} E_t[\tilde{d}_{t+\tau}]$$

(A2) Owners' equity accounting, in two distinct parts

$$(A2a) \quad y_t = y_{t-1} + x_t - d_t \qquad (A2b) \quad \frac{\partial y_t}{\partial d_t} = -1, \quad \frac{\partial x_t}{\partial d_t} = 0$$

Clean surplus (A2a) does not imply (A2b): dividends reduce book value, not current earnings, by convention.

(A3) Linear information dynamics

Abnormal earnings follow a modified autoregressive process (slide 5). Assumptions (A1) and (A2a) alone already yield the residual income identity; (A3) delivers the closed form.

Clean surplus converts the dividend stream into accounting numbers

Define abnormal earnings and solve clean surplus for dividends:

$$x_t^a \equiv x_t - (R_f - 1) y_{t-1} \implies d_t = x_t^a - y_t + R_f y_{t-1}$$

Substituting into (A1), with $E_t[\tilde{y}_{t+\tau}]/R_f^\tau \rightarrow 0$, yields residual income valuation:

$$P_t = y_t + \sum_{\tau=1}^{\infty} R_f^{-\tau} E_t[\tilde{x}_{t+\tau}^a] \quad (1)$$

“this formula is peculiar because one interprets it by referring to accounting concepts, yet the formula works regardless of the accounting principles that measure book values and earnings.”

Ohlson (1995, p. 667)

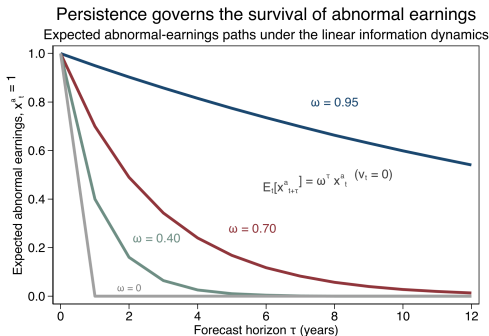
The information dynamics impose mean reversion on abnormal earnings

$$\tilde{x}_{t+1}^a = \omega x_t^a + v_t + \tilde{\varepsilon}_{1,t+1}$$

$$\tilde{v}_{t+1} = \gamma v_t + \tilde{\varepsilon}_{2,t+1}$$

with $0 \leq \omega, \gamma < 1$. The variable v_t , other information, captures “value relevant events that have yet to have an impact on the financial statements” (p. 668).

Competition erodes abnormal earnings; the accounting eventually records what the market knows.



Price is a persistence-weighted average of two polar valuation models

The closed form (eq. 5):

$$P_t = y_t + \alpha_1 x_t^a + \alpha_2 v_t$$

$$\alpha_1 = \omega / (R_f - \omega),$$

$$\alpha_2 = R_f / [(R_f - \omega)(R_f - \gamma)]$$

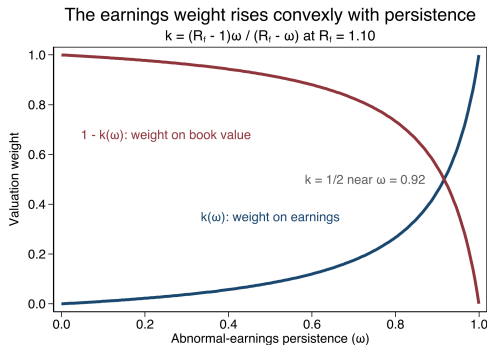
Restated as a weighted average
(eq. 7):

$$P_t = k(\varphi x_t - d_t) + (1 - k)y_t + \alpha_2 v_t$$

$$\varphi = R_f / (R_f - 1),$$

$$k = (R_f - 1)\omega / (R_f - \omega)$$

$$\omega = 1: P_t = \varphi x_t - d_t \quad \omega = 0: P_t = y_t$$



Dividends reduce book value, not current earnings

A dividend removes a dollar of assets: book value falls; current earnings do not (A2b). Price falls by that dollar, and expected earnings by the forgone interest:

$$\partial P_t / \partial d_t = -1 \quad (\text{P3}) \qquad \partial E_t[\tilde{x}_{t+1}] / \partial d_t = -(R_f - 1) \quad (\text{P1})$$

“Dividends reduce market value on a dollar-for-dollar basis because dividends (i) reduce book value similarly on a dollar-for-dollar basis but (ii) do not affect the expected abnormal earnings sequence . . .”

Ohlson (1995, p. 681)

Clean surplus alone does not suffice: a system charging dividends against current earnings also satisfies clean surplus, yet displacement fails (p. 673). **The Modigliani-Miller property is derived from accounting conventions, not assumed.**

Value-relevance regressions estimate special cases of the valuation function

Valuation model	Levels specification	Differenced (returns) form
Earnings capitalization, $P = \varphi x$	$MV = \alpha + \beta_1 E$	R_t on $\Delta E / MV_{t-1}$
Book value, $P = y$	$MV = \alpha + \beta_2 BV$	R_t on E / MV_{t-1} ($\Delta BV = E - D$)
Ohlson eq. (7)	$MV = \alpha + \beta_1 E + \beta_2 BV$	R_t on $\Delta E / MV_{t-1}$ and E / MV_{t-1}

Differencing the combined levels model delivers the Easton and Harris (1991) returns specification with both earnings changes and earnings levels.

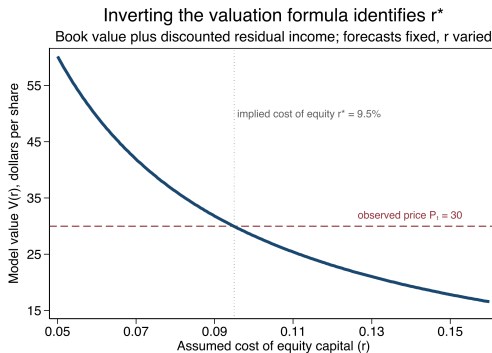
The theory disciplines the regression: the coefficients are functions of (ω, R_f) , the earnings and book-value weights move against each other as persistence varies, and v_t is a named omitted variable rather than an error term.

Botosan (1997) inverts the valuation function to measure the cost of equity

The design: 122 machinery firms; a hand-constructed disclosure index from 1990 annual reports; the residual income formula with analyst-based forecasts; observed price substituted for P_t ; solve for the internal rate r .

The finding: greater disclosure is associated with a lower implied cost of equity for firms with low analyst following; she fails to find the association for heavily followed firms.

Her reference list cites Ohlson (1995) directly.



Successive implementations standardized the inversion

Paper	Contribution to the inversion
Gebhardt, Lee, and Swaminathan (2001, <i>JAR</i>)	Standardized three-stage implementation; ROE fades to the industry median
Claus and Thomas (2001, <i>JF</i>)	Market-level inversion; implied equity premium near three percent
Easton (2004, <i>TAR</i>)	PEG-based estimators from two years of forecasted earnings
Botosan and Plumlee (2005, <i>TAR</i>)	Construct validity: only some proxies associate with risk as predicted
Hou, van Dijk, and Zhang (2012, <i>JAE</i>)	Model-based earnings forecasts replace analyst forecasts

The valuation-model inversion survives across fifteen years of implementations, from Botosan (1997) to Hou, van Dijk, and Zhang (2012); only the forecast input and the terminal-value assumptions change.

The model was canonized while its distinctive content went largely untested

Dechow, Hutton, and Sloan (1999, *JAE*)

Estimating the information dynamics and imposing them on the valuation improves only modestly over simple capitalization of analyst forecasts.

Lo and Lys (2000, *JAAF*)

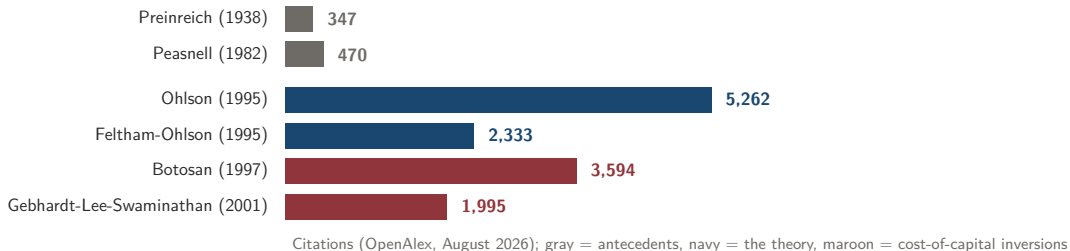
Most papers citing Ohlson (1995) implement the residual income identity or levels regressions while ignoring the dynamics and the other-information variable entirely.

Ohlson (2001, *CAR*)

The author's own corrective: v_t is forecastable next-period earnings news; empirical work that discards it is not testing the model.

A theory can be institutionalized by citation while its distinctive content goes untested.

One theory paper anchored two empirical literatures



AAA Notable Contributions to Accounting Literature Award, 2000; Seminal Contributions to Accounting Literature Award, 2013; Accounting Hall of Fame, 2015.

Summary

- From the present value of expected dividends and clean surplus alone, price equals book value plus the present value of expected abnormal earnings, under any accounting principles satisfying clean surplus; the identity predates the model by decades.
- The information dynamics convert the identity into a closed-form valuation function in which earnings and book value are complementary indicators weighted by a single persistence parameter, and dividend displacement follows from accounting conventions rather than from assumption.
- Read forward, the model disciplines value-relevance regressions; inverted, it measures the implied cost of equity capital. The empirical literature it created, however, largely set aside the information dynamics that constitute its distinctive contribution.

Ohlson (1995) · Botosan (1997) · Dechow, Hutton, and Sloan (1999) ·
github.com/sbb151/acctg597a-research-design